



Timely deployment of contaminant warning systems requires on-line sensors and advancement of data analysis and decision support systems to accurately detect water quality changes. As a demonstration of event detection in water quality data, three water quality change-detection algorithms were developed and used to detect changes in water quality observed at four locations within a distribution system. Each data set was “spiked” with simulated anomalous water quality values of 1 h duration and 10 levels of spike strength. The receiver operating characteristic (ROC) curve is proposed as an objective means of assessing and comparing water quality change-detection algorithm results that is also able to provide the tradeoff between missed detections and false alarms on a single plot. The area under the ROC curve provides a measure of the sensitivity and specificity of each algorithm for each spike strength and location. Resulting areas under the ROC curve range from 0.46 for spike strengths of 1.0 (background) to 0.98 for strengths of 3.5 standard deviations from the mean, where an ROC curve area of 1.0 indicates perfect detection.

Detecting Changes in Water Quality Data

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Protection of drinking water systems from accidental and malevolent contamination events has taken on increased importance in recent years because of heightened security concerns (Roberson & Morley, 2005). A number of data sources, including customer complaints and public health monitoring, can be used to identify contamination in water distribution systems (Roberson & Morley, 2005). However, the primary data source for detecting contamination events is a contaminant warning system (CWS) composed of on-line sensors, connected supervisory control and data acquisition (SCADA) systems, data analysis algorithms, and decision-support tools (Hasan et al, 2004; Grayman et al, 2001). An important component of any CWS is the ability to rapidly and accurately detect a significant change in water quality as it occurs within a distribution system. In the absence of contaminant-specific on-line sensors, detection of a significant change must depend on the signals generated by existing on-line water quality sensors and the proper analysis of these signals. At its essence, change detection is a way to separate anomalous conditions from normal background operating conditions. Detection of a change does not necessarily imply that the cause of that change has also been identified.

To date, two general approaches to developing and testing event detection using water quality signals have been examined. First, laboratory and test-

loop evaluation of sensors and associated event detection algorithms provides direct measurement of chemical changes in background water quality caused by specific contaminants (Hall et al, 2007; Kroll & King, 2006; Byer & Carlson, 2005; Cook et al, 2005). Results from these physical experiments provide the deviations in background water quality signals that are indicative of contamination events; these responses are being integrated into event detection methods (Yang et al, 2007; Kroll & King, 2006). A drawback of the laboratory and test-loop results and the resulting algorithms is that variation of the background water quality in these systems may be considerably less than the variation observed in actual distribution systems.

A second approach to event detection is based on signal processing and pattern recognition techniques borrowed from other fields. This approach uses background water quality observations as measured by sensors deployed in distribution systems with numerically simulated events inserted into these observations as test data (Cook et al, 2006; Klise & McKenna, 2006; McKenna et al, 2006). This second testing approach has the advantage of employing actual background variation in water quality; however, it does not necessarily provide accurate water quality sensor response to a particular contaminant. This second approach was used to examine three signal-processing-based change-detection algorithms and to discuss a generally applicable means of evaluating these algorithms.

BACKGROUND INFORMATION

Three approaches to water quality change detection are evaluated. Each approach is based on a comparison between predicted water quality at the next time step in the future and the actual water quality measurement made at that time step. A significant difference between predicted and measured water quality indicates an anomalous water quality event. This approach relies on two capabilities: accurate predictions of water quality at future time steps and a working definition of “significant.”

Predicting future values of time series data is well-developed in many diverse fields including surface water hydrology (Bras & Rodriguez-Iturbe, 1993) and finance (Campbell et al, 1997). Although many data-driven approaches to time series prediction have been developed, all require some measure of the past behavior of the time series and generally require that the time series exhibit temporal autocorrelation such that the most recently measured values provide information that can be exploited to predict future values. All distribution system

water quality data sets examined to date display strong temporal autocorrelation.

Autocorrelation in water quality time series quantifies the similarity of water quality measurements that are taken close together in time. At short time lags, comparison of any two measurements will show them to be relatively similar or strongly correlated. As the time lag between measurements increases, the correlation between

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measurements decreases until at some separation in time, the measurements become independent. Strong autocorrelation allows for accurate prediction of future water quality values based on recent values. In most distribution systems, the nature of this autocorrelation is influenced by diurnal patterns that reflect utility operations over the time scale of a single day as well as longer-term seasonal trends that affect water sources.

The amount of deviation between predicted and observed values that constitutes a significant change in water quality is a relative measure: lower tolerances for change or higher sensitivity lead to more conservative decision-making and vice versa. Evaluation of change detection requires that the results be examined on a common scale to assess the tradeoffs between false-positive (FP) and false-negative (FN) decisions as a function of the sensitivity of the change-detection algorithm. This evaluation must remain independent of the basis of the algorithm and the form and quantity of the input data. Here, the receiver operating characteristic (ROC) curve is proposed as an evaluation tool. ROC curves were originally developed to evaluate the ability of radar systems to detect aircraft during World War II (Lusted, 1971). The curves have since been applied in a wide range of fields in which separation of signal from noise is the ultimate goal, including effectiveness of psychological treatments (Swets, 1988), evaluation of medical diagnoses and treatments (Collinson, 1998; Metz, 1978), comparison of geophysical sensors designed to detect unexploded ordnance and land mines (Carin et al, 2002; Barrow & Nelson, 2001; Gao & Collins, 2000; Gelenbe & Kocak, 2000), and comparison of predictive techniques for surface water quality (Morrison et al, 2003). ROC curves provide an objective means of evaluating water quality change-detection algorithms in water distribution systems; this approach is demonstrated in this article.

CHANGE-DETECTION ALGORITHMS

Three approaches to detecting changes in water quality data as measured using online sensors within water distribution systems are evaluated using ROC curves—time series increments (INC), multivariate (MV) distance, and linear filters (LF). The relationship between the false alarm rate (FAR) and the probability of event detection for each change-detection approach is evaluated by constructing ROC curves. This is a general approach that can be used to evaluate any change-detection system.

Data standardization. It is assumed that at any monitoring location within a utility, multiple water quality signals are recorded and transmitted to a central control center through a SCADA system. The different water quality sensors at that location may take different amounts of time to obtain a measurement, for example, a few seconds for pH or a few minutes for total organic carbon. However, it is assumed that the time at which the measurement data are received at the control center is the time of the measurement. Each water quality signal must be reported simultaneously for a given sensor location. The polling interval of the SCADA system (i.e., every few minutes to every hour) is the water quality measurement interval and is assumed to be held constant for all sensors at a given location across the length of the monitoring period.

The data collected from the different water quality sensors are in different units (e.g., mg/L, pH, ntu) and therefore need to be standardized to a common scale. Here, standardization is achieved by subtracting the data set mean, μ , from each water quality measurement, $X(t)$, and then dividing this difference by the standard deviation, σ , of the data set:

$$Z(t) = \frac{X(t) - \mu}{\sigma} \quad (1)$$

The standardized data, $Z(t)$, are now in units of standard deviations away from the mean. Typical standardized values for the data sets with multiple sensor types range from 0.0 ± 3.5 , with occasional excursions beyond ± 5.0 . The mean and standard deviation are calculated within a moving window of previous measurements of length P . As new measurements arrive, the mean and standard deviation are recalculated using the most recent P measurements.

Time series increments. The simplest approach for identifying changes is that of time series increments, or differences between successive measures of a water quality variable. These types of differences are implicitly a prediction model in which the prediction of the standardized water quality parameter value at the current time step, $Z^*(t)$, is simply the standardized value measured at the previous time step:

$$Z^*(t) = Z(t-1) \quad (2)$$

This dependence on only the single previous measurement is the definition of a Markov model (Taylor & Karlin, 1998). The time series increments, $\delta(t)$, are calculated as

$$\delta(t) = Z^*(t) - Z(t) = Z(t-1) - Z(t) \quad (3)$$

These differences are calculated on standardized data so that the units of the δ s are also in units of standard deviation. The deviations beyond a prescribed threshold in standardized space, δ_c , for example, two standard deviations, are identified as anomalous.

Operationally, the window length P is selected, and the mean and standard deviation of the P values are calculated for use in standardization as described previously. The goal of this standardization is to obtain the standardized value of $Z(t-1)$ to use as a prediction of $Z(t)$. When $Z(t)$ becomes available, it is also standardized using the same mean and standard deviation of the raw data contained in the window of previous values. A large enough P must be chosen to provide representative values of the mean and standard deviation while not being so large as to overly smooth the data. The difference between $Z^*(t)$ and $Z(t)$ is then compared against the threshold δ_c to determine if an event has been detected. The window is moved one time step ahead to include $Z(t)$, and the standardized values are recalculated. The length of the standardization window is the only adjustable parameter in this detection algorithm.

The predictions and differences within the standardized space can be created for any specific water quality signal (e.g., chlorine or pH). Here, each water quality signal is treated separately, and then all water quality signals are combined by taking the average absolute value of the differences in units of standard deviation across all signals and then comparing these average differences to δ_c .

Multivariate distance. The MV distance algorithm considers changes in water quality based on the difference between successive water quality measurements in a multivariate space defined by the water quality signals. For example, a four-dimensional space may be defined by the standardized values of pH, chlorine, specific conductivity, and oxidation reduction potential. The distance measure used to distinguish between normal and anomalous water quality is based on the Euclidean distance from the standardized water quality measurement at the current time step to the mean water quality value over the previous P time steps. The current water quality and the mean water quality are standardized and therefore a function of the size of the moving window, P . A constant distance threshold, δ_c , is applied to the distance measure to determine if the water quality measurement is anomalous. Distances above the threshold are categorized as anomalous, and distances below the threshold are considered normal.

The distance measure, δ , used in the MV algorithm is the difference between the Euclidean distances of successive points to the mean in n dimensional space:

$$\delta = \sqrt{\sum_{j=1}^n [Z(t)_j - Z(m)_j]^2} - \sqrt{\sum_{j=1}^n [Z(t-1)_j - Z(m)_j]^2} \quad (4)$$

in which n is the number of water quality parameters, $Z(t)_j$ is the j^{th} water quality parameter in the current time step, $Z(t-1)_j$ is the j^{th} water quality parameter in the previous time step, and $Z(m)_j$ is the mean value of the j^{th} water quality measurement. The measured multivariate distance is then compared to the threshold δ_c to determine if it is an anomalous change in water quality. The MV algorithm distinguishes water quality anomalies when the distance to the mean water quality value is significantly different from the distance to the mean water quality value of the previous time step. For the MV algorithm, only a combined, multivariate change is calculated; results for individual sensors are not. Recent work on the MV algorithm (Klise & McKenna, 2006) resulted in development of a variation on the approach described here.

Linear filters. A linear predictor estimates the current value of a time series as a linear combination of previous samples. In its most general form, this approach is also known as the autoregressive (AR) moving average predictor (Bras & Rodriguez-Iturbe, 1993), and the most common representation is

$$Z^*(t) = -\sum_{i=1}^P a_i Z(P-i) \quad (5)$$

in which $Z^*(t)$ is the predicted water quality value, $Z(P-i)$ is the previous observed values within the standardization window, a_i is the predictor coefficients, and P is the order of the prediction filter polynomial. P is also the size of the standardization window as described previously. The error generated by this estimate is

$$\delta(t) = Z^*(t) - Z(t) \quad (6)$$

in which $Z(t)$ is the observed true signal value. The difference $\delta(t)$ is compared to the threshold δ_c to determine whether the difference is significant. Several methods are available to solve for the values of a_i . For this implementation, the autocorrelation method of AR modeling was used to find predictor coefficients that produce an unbiased and minimum variance estimate of the unknown true value (Bras & Rodriguez-Iturbe, 1993).

DATA SETS

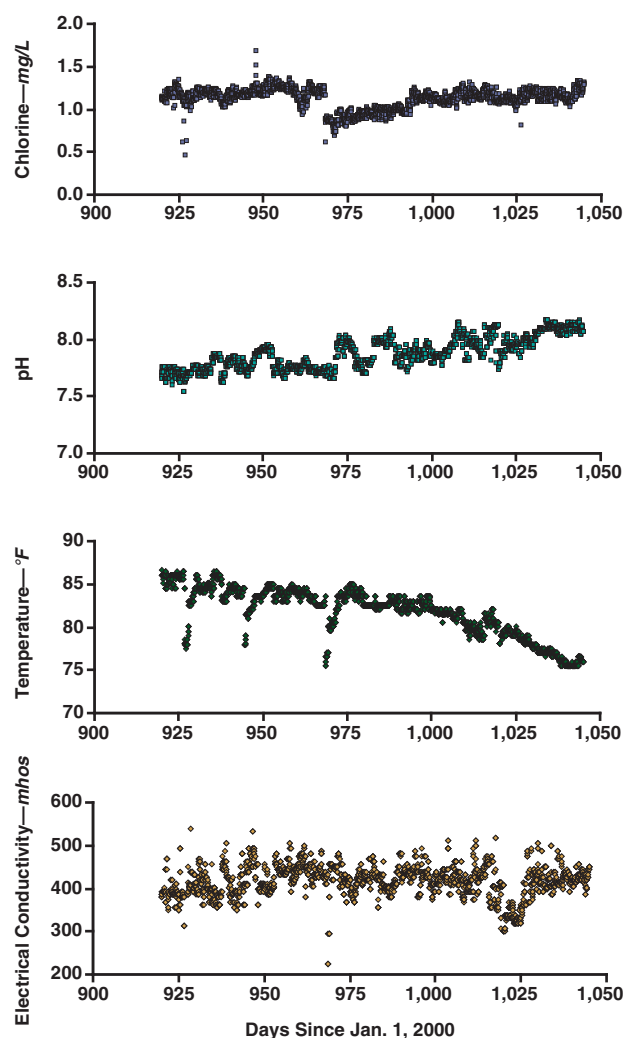
Rigorous assessment of event-detection algorithms requires a set of known events against which these algorithms can be evaluated. However, water quality data sets from operating utilities that contain numerous events of anomalous water quality, where the location, timing, and source responsible for each change are all well documented, do not exist. Water quality data sets collected at four locations in a moderate- to large-sized network (between 0.5 million and 1 million customers) in the United States were used to represent background water

quality conditions. Simulated anomalous water quality events were then added to these data.

Measured water quality data. Water quality sensor data from four locations (P2, P7, P8, and P9) in the network were examined, and each location was analyzed independently. These locations were chosen to provide samples from different portions of the network. At each location, four water quality signals—free chlorine, pH, temperature, and electrical conductivity (EC)—were monitored simultaneously with a 1-h sampling interval for 3,000 h (125 days). The beginning time of each data set studied varied from location to location to provide water quality data at different times of the year.

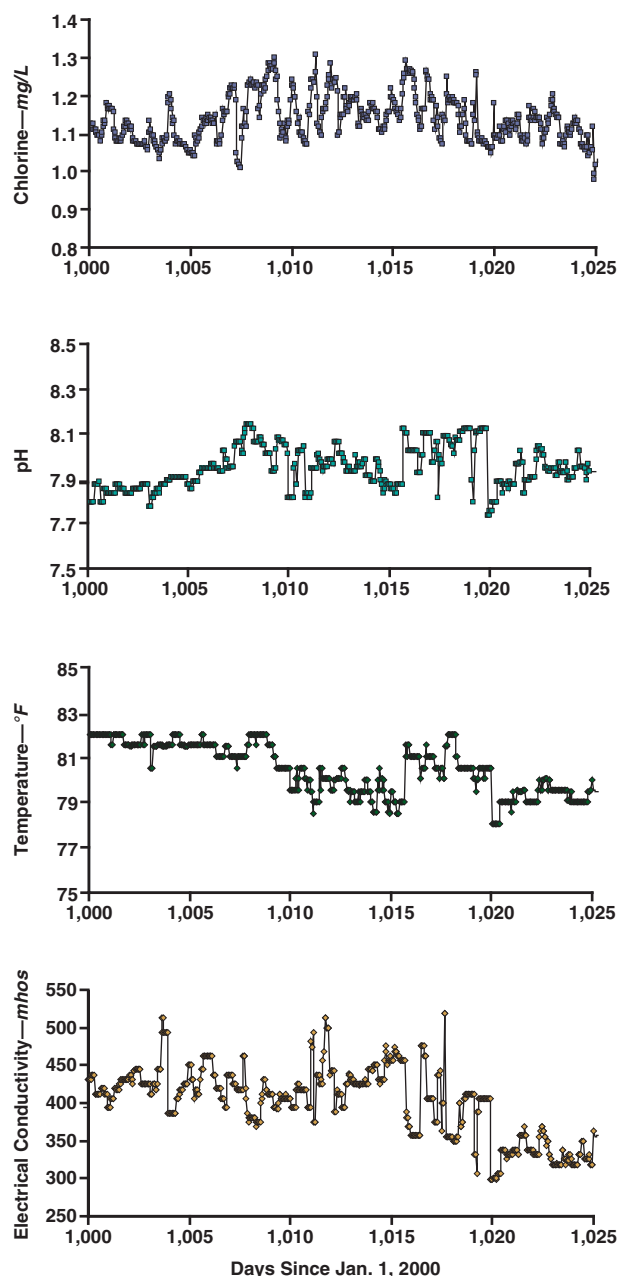
As an example, the time series of the four water quality signals measured at location P2 are shown in Figure 1. These data exhibit typical noise levels seen in data col-

FIGURE 1 Example water quality data sets for location P2



3,000 values are shown in each plot.

FIGURE 2 Expanded view of 25 days (625 h) of water quality measurements for location P2



lected from a distribution system as well as short-term deviations and longer-term trends. The chlorine data range from a few measurements less than 0.5 mg/L to a few that are greater than 1.5 mg/L. Significant short-term (i.e., a few days or less) deviations in chlorine, temperature, and EC may be related to periods of sensor calibration, abnormal mixing of source waters, or noise in the SCADA system. The increasing trend in pH and the decreasing trend in temperature, which persist through-

out the 125-day monitoring period, are the result of seasonal changes. Shorter-term cycles can also be seen on top of these longer-term trends (Figure 1).

The different amounts and types of variation in the water quality time series can be better appreciated by examining a single subset of the data. Data from a 25-day (625-h) period (1,000–1,025 days after Jan. 1, 2000), are shown in Figure 2. Note that both the x and y axes of Figure 2 have been expanded relative to Figure 1. The details in Figure 2 show a relatively strong diurnal cycle for chlorine and a much weaker diurnal cycle for EC that is masked by the relatively high level of noise in the EC sensor. The temperature sensor appears to have a resolution of 0.4° that leads to the stepped appearance of this signal. The lines connecting adjacent data points are near vertical when there is a large change between consecutive measurements. Examination of Figure 2 indicates that the largest deviations between consecutive measurements are 20–30% for chlorine and EC data and 2–3% for pH and temperature data. Monitoring environments with changes of 20–30% between consecutive measurements of background water quality can make change detection challenging.

Event simulation. To add each anomalous event, or spike, to the water quality data, the mean and standard deviation of all 3,000 h of data are calculated for each signal at each location. Then each data stream (Eq 1) is standardized to achieve a mean of 0.0 and a standard deviation of 1.0. Anomalous event times are determined randomly such that each time step in the last 2,000 h of the data series has an equal probability of being selected. Events are not located in the first 1,000 h of data because these data are reserved to be representative of the background water quality and for development of the standardized background data values prior to the algorithms encountering any events.

Each simulated event has a length of a single time step, 1 h. At each time selected for simulation of an event, each existing, standardized water quality value is multiplied by the same spike strength. Spike strength values are in units of standard deviation and range from 1.0 (no change in the water quality) to 3.5 (a change of 3.5 standard deviations away from the mean). This approach is dependent on the background value of the water quality for each sensor at the randomly chosen time step. For example, if the standardized chlorine value at a chosen time step is –1.5 and it is multiplied by a spike strength of 2.0, the spiked chlorine value becomes –3.0, or 3 standard deviations below the average chlorine value. At the same time step, the standardized value of pH may be 0.1, and after application of the spike of 2.0, it is only 0.2 standard deviations away from the mean pH value. This approach to adding events to the water quality data does not attempt to reproduce any specific chemical reactions that might increase or decrease specific water quality parameters when combined with a particular contami-

nant. However, the approach does mimic the general behavior of experiments when one or more water quality values increase when a contaminant is introduced while others may decrease or stay relatively constant (e.g., Hall et al, 2007).

The final step in simulating a water quality event is to transform the spiked signals back from the standardized space to the original raw unit space. This is done using the reverse of the standardization procedure with the original mean and standard deviation of the data as calculated prior to the addition of the events.

EVALUATION APPROACH

The accuracy of a change-detection algorithm is assessed by its ability to place the current state of water quality at a location within the distribution system into one of two classes: background or anomalous. Accuracy is composed of the ability of the algorithm to detect the anomalous operating conditions (this is the sensitivity of the algorithm) and to correctly exclude the normal operating conditions from being inadvertently classified as anomalous (this is the specificity of the algorithm). Sensitivity and specificity are inextricably linked—as the sensitivity increases, more anomalous events are detected, but at the cost of decreasing the specificity as more and more normal conditions become classified as anomalous. In other words, as the sensitivity is increased, the number of FP events must also increase. Decreasing the sensitivity of the algorithm results in missing some of the anomalous events and thus increases the number of FN events.

A correct result or one of two types of errors—FP or FN—occurs when the result of a change-detection algorithm is compared with the true result. Across application of the change-detection algorithm at multiple time steps, the proportion of the correct, FP, and FN decisions will be a function of the threshold, δ_c , at which the discrimi-

nation between normal and anomalous is determined. It is necessary to examine the tradeoff between the different types of results as a function of changes in the threshold in order to determine the threshold at which to operate the algorithm and to understand what proportion of anomalous events will go undetected and what proportion of events that are identified will turn out to be FPs.

Figure 3 provides a simple example of a threshold being set to discriminate between two populations of a water quality signal. For example, these could be populations of total dissolved solids (TDS) values, with the two populations caused by varying proportions of water from two sources reaching the sensor. A water quality event, the anomalous condition, or a true positive (TP), is caused when the pump at one source does not operate

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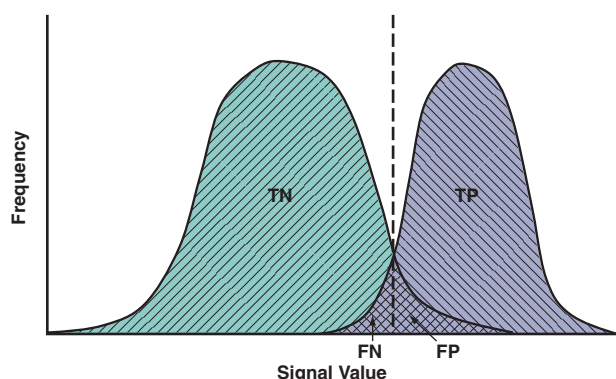
correctly and too much of the water with higher TDS reaches the sensor, creating the measurements in the higher TDS distribution to the right side of Figure 3. This event is identified when the measured TDS value exceeds the threshold (Figure 3). The goal is to set the threshold such that every time the pump does not function correctly, i.e., the threshold is exceeded, and when the pump is functioning correctly, i.e., the normal operating condition, or the true negative (TN), the measured TDS is below the threshold. It is difficult to choose the correct setting because the two TDS distributions overlap such that the same TDS value can occur under normal or anomalous conditions. For instance, at the current threshold level, the high end of the TN population will exceed the threshold, and these measurements will be falsely classified as anomalous events—the FPs. Similarly, values in the low

TABLE 1 Relationships between actual and estimated values

Condition	True Background Condition	True Anomalous Condition
Estimated anomalous condition	FP	TP
Estimated background condition	TN	FN
Column Sum	FP + TN	TP + FN
	Specificity = $TN / (FP + TN)$	Sensitivity = $TP / (TP + FN)$
	FAR = $FP / (FP + TN)$	
	Specificity = $1 - FAR$	Sensitivity = PD

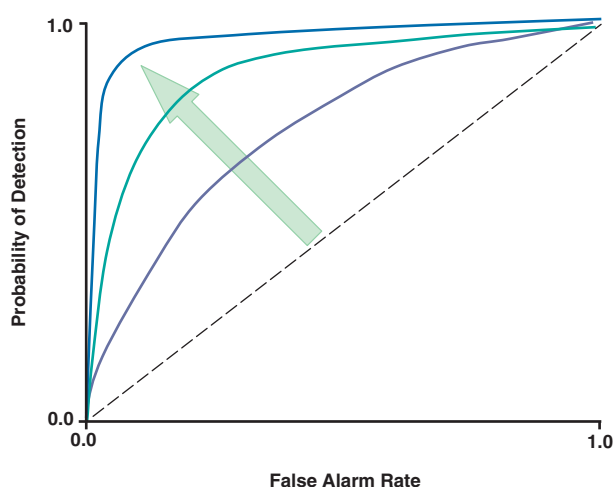
FAR—false alarm rate, FN—false negative, FP—false positive, PD—probability of detection, TN—true negative, TP—true positive

FIGURE 3 Example of two populations of water quality values



FN—false negative, FP—false positive, TN—true negative, TP—true positive

FIGURE 4 Schematic diagram of four receiver operating characteristic curves



end of the TP distribution will be falsely classified as negative events, or FNs. For any threshold setting, there will be a tradeoff between the number of FP and FN classifications. Moving the threshold to the left (i.e., lower values) increases the sensitivity of the discrimination and will result in fewer FNs but at the expense of increasing the number of FPs. The relation between the estimated and actual values and the different types of errors are shown in Table 1.

The ROC curve (Figure 4) defines the probability of detection (PD) that can be obtained as a function of the corresponding FAR, in which the FAR is the number of FPs divided by the total number of values that are actually below the threshold (TN + FP; Table 1). The PD is defined as the number of TPs divided by all events that

actually exceed the threshold (TP + FN; Table 1). For a perfect change-event-detection algorithm, at all threshold settings the FAR would be 0.0, and the PD value would be 1.0. For this case of perfect discrimination, the ROC curve is composed of a vertical line along the left edge of the graph and a horizontal line across the top of the graph—the two lines would meet at the upper left corner of the graph. The area under this ROC curve, representing perfect event discrimination, would be 1.0. An algorithm that provides no useful information on events would produce an ROC curve from the lower left to upper right corners of the graph (dashed line, Figure 4) with the area under the curve equal to 0.5. This 45° line is the worst possible case because if the ROC curve goes below this line, it would be possible to simply use the opposite of the result produced by the algorithm to improve the decisions (Selin, 1965).

At high values of the threshold (e.g., toward the right edge of the graph in Figure 3), any event detected will be a TP and there will be no FPs (bottom left corner of Figure 4). As the threshold is lowered, the algorithm becomes more sensitive and the proportion of anomalous events detected (TPs) increases, but so does the number of FPs, pushing the curve up and to the right on the FAR axis (Figure 4). As the threshold is decreased further, both the number of true events detected and the number of FPs increase. The slope of the ROC curve is the increase in TPs per increase in FPs. It is the slope of this increase that defines the accuracy of the change-detection algorithm. Figure 4 shows a series of ROC curves representing improved performance (larger areas under the curve) as the curves approach the upper left corner of the graph. The area under the ROC curve is the preferred single-valued measure of accuracy of the technique being evaluated (Swets, 1988). The area under the curve can range from 0.5 to 1.0. However, this area is a nonunique indicator of performance because differently shaped curves can have the same area but different PD values for the same FAR value.

Development of the ROC curve for a particular location allows a distribution system analyst to set the threshold of the detection algorithm to achieve an acceptable PD. As the curves in Figure 4 show, the more conservative the setting of this threshold, the higher the resulting FAR. This curve provides the analyst with information on how often FPs can be expected for a given level of event detection. Obviously, to detect most events, it will be necessary to investigate a large number of FPs.

In a practical setting, given an event-detection algorithm, the steps to developing an ROC curve for a monitoring location are as follows:

- Collect baseline water quality data at that location for a period of time.
- Seed the baseline data with events. These events can be relatively simple spikes that last for a single time step, as done here, or more complicated events based on the chemistry of a particular contaminant(s) and modeled

reactions between that contaminant and constituents of the baseline water.

- Vary the threshold in the event-detection algorithm while recording the FPs and FNs to create the ROC curve.

These steps are demonstrated in the next section. Once the curve is constructed, it is necessary to select a point on the curve where the PD is maximized and the FAR is minimized, typically the upper-left point on the curve. The

the data set. At each of these points in time, a spike was added to the background water quality as described previously. Ten spike strengths were evaluated. For each spike strength, an ROC curve was calculated across the 107 events by varying the threshold value δ_c .

Parameters defining the operation of the different change-detection algorithms were set for use on these data sets. The main parameter that must be set for each

algorithm is the length of the moving window of previous time steps, P , used for the data standardization. The approach to setting the value used for this study was to select a single spike strength, 1.5 for these analyses, and determine the average ROC curve area across all four monitoring locations as a function of P .

This testing was done separately for each algorithm. For the INC algorithm, several values of the moving window length, P , ranging from 50 to 150 h, were used, and there were only small changes in the results across this range of values. A P value of 120 h (5 days) was used for all analyses with the INC algorithm. The optimal value of P for the MV algorithm was 250 h. This P value is the smallest window that resulted in the largest average area under the ROC curves for the four locations. For the LF algorithm, P values ranging from 12 to 168 h (7 days) were examined, and a

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event-detection algorithm is then used with a threshold corresponding to this point. The utility analyst is then made aware of the possibility that a water quality event will be missed (i.e., an FN) and of how frequently false alarms can be expected.

RESULTS

The three algorithms were applied to the four monitoring locations. At each monitoring location, 107 points in time were randomly selected from the last 2,000 h of

separately for each algorithm. For the INC algorithm, several values of the moving window length, P , ranging from 50 to 150 h, were used, and there were only small changes in the results across this range of values. A P value of 120 h (5 days) was used for all analyses with the INC algorithm. The optimal value of P for the MV algorithm was 250 h. This P value is the smallest window that resulted in the largest average area under the ROC curves for the four locations. For the LF algorithm, P values ranging from 12 to 168 h (7 days) were examined, and a

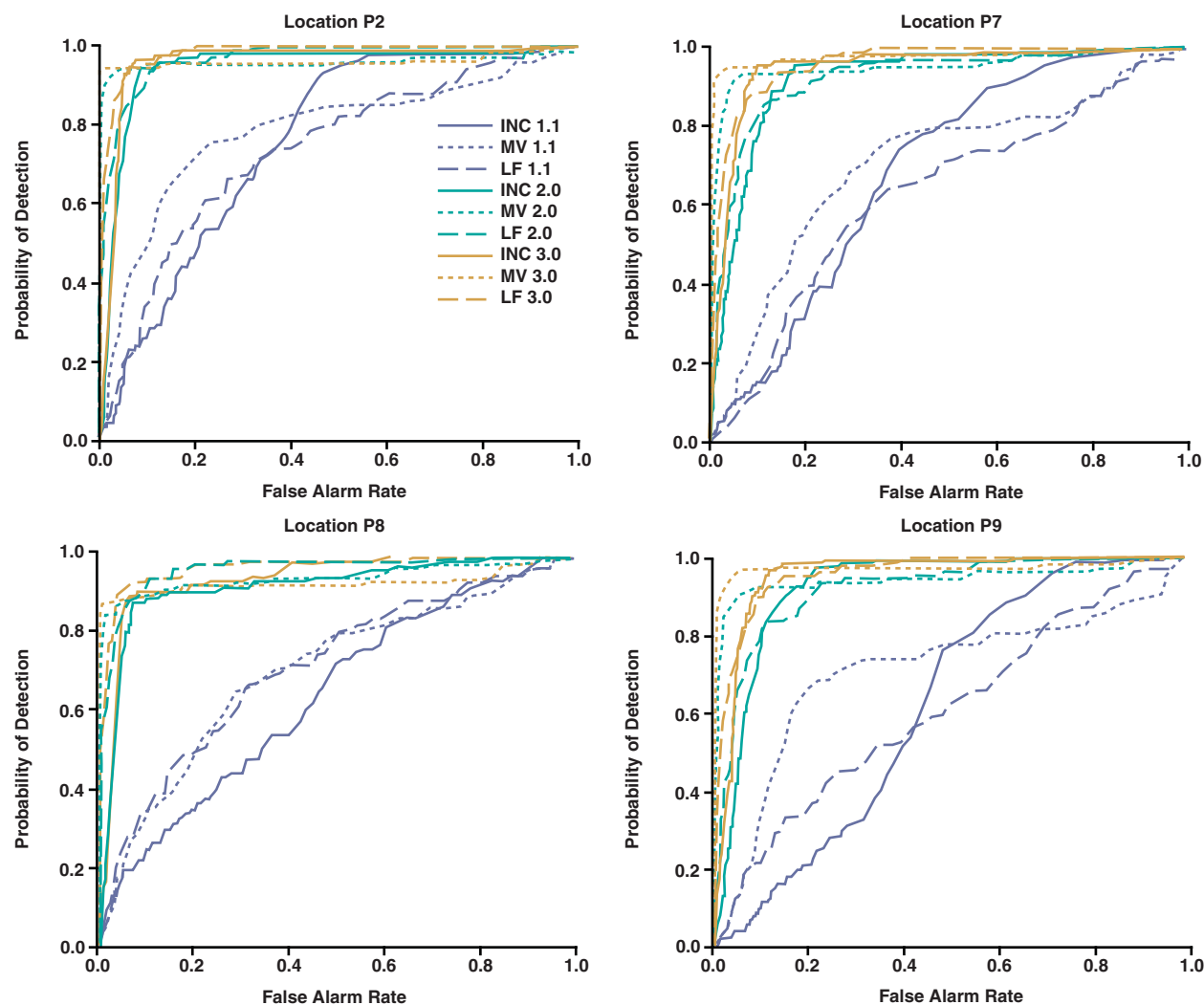
TABLE 2 Areas under the combined, four sensors, ROC curve for each algorithm at each location for all 10 spike strengths

Spike Strength	Location											
	P2			P7			P8			P9		
	Parameter											
	INC	MV	LF	INC	MV	LF	INC	MV	LF	INC	MV	LF
1.0	0.55	0.55	0.56*	0.54*	0.53	0.50	0.46	0.49	0.50*	0.46	0.49	0.51*
1.1	0.76	0.78*	0.73	0.68	0.70*	0.63	0.64	0.70	0.72*	0.61	0.71*	0.61
1.2	0.83	0.85*	0.81	0.76	0.78*	0.73	0.77	0.83	0.84*	0.73	0.79*	0.70
1.3	0.87	0.90*	0.87	0.82	0.83*	0.80	0.83	0.89	0.90*	0.79	0.83*	0.77
1.4	0.90	0.93*	0.90	0.85	0.87*	0.84	0.87	0.91	0.93*	0.83	0.86*	0.81
1.5	0.92	0.94*	0.92	0.87	0.91*	0.86	0.89	0.93	0.95*	0.86	0.89*	0.84
2.0	0.94	0.96	0.97*	0.92	0.95*	0.92	0.92	0.95	0.97*	0.92	0.94*	0.92
2.5	0.95	0.96	0.98*	0.94	0.97*	0.95	0.93	0.94	0.97*	0.94	0.96*	0.95
3.0	0.95	0.96	0.98*	0.94	0.98*	0.96	0.94	0.93	0.97*	0.95	0.97*	0.96
3.5	0.95	0.96	0.98*	0.95	0.97*	0.97*	0.94	0.93	0.97*	0.95	0.97*	0.97*

INC—time series increments, LF—linear filters, MV—multivariate distance, ROC—receiver operating characteristic

*The largest area under the ROC curve at each location across the three algorithms.

FIGURE 5 Receiver operating characteristic curves resulting from application of the three change-detection algorithms at each location for water quality data with three spike strengths



INC—incremental, LF—linear filters, MV—multivariate

P value equal to 24 h produced the best results. It is possible to determine a separate P value for each site. However, here the P value is determined as the average best value across all four sites. Certainly a separate P value could be determined for each monitoring location, and these may improve the results. In a field setting in which large numbers of known events do not exist, this same approach of using simulated water quality events on top of previously recorded background water quality to set P could be used. Alternatively, identification of the P value that minimizes the average prediction error [$\delta(t)$ in Eq 6] in the background water quality could be used.

The four locations, 10 spike strengths of the simulated events, and different algorithms resulted in creation of a large number of ROC curves. Only the ROC curves calculated on the combination of all four water quality sig-

nals at each of the four locations for three spike strengths: 1.1, 2.0, and 3.0 σ are shown (Figure 5). The spike strength of 1.1 results demonstrated ROC curves for a very low signal-to-noise ratio, whereas the resulting ROC curves for spike strengths of both 2.0 and 3.0 show much clearer separation of anomalous events from the background signal.

There is a large difference in the ROC curves between a spike strength of 1.1 and spike strengths of 2.0 or 3.0. Also, there is considerable overlap in the ROC curves for spike strengths of 2.0 and 3.0, indicating that above a certain spike strength, the change-detection algorithms produce similar results. Figure 5 indicates that the ROC curves furthest from the upper left corner of the graph occur at location P8. Across all locations it is difficult to determine which, if any, of the three algorithms produces the best

results. The differences between algorithms and locations are examined further by calculating the area under each curve and the FAR value for a single PD value of 0.95.

The areas under all ROC curves for all locations and spike strengths are shown in Table 2. Across the three algorithms, the largest area under the ROC curve at each location is indicated. As might be expected, the simplest algorithm, the time-series increment approach, consistently produces ROC curves with the smallest areas when compared with the other algorithms. The multivariate algorithm gives the best performance at locations P7 and P9, whereas the linear filter algorithm produces the largest areas under the ROC curves at location P8. Results are mixed at location P2. The ROC areas increase most quickly as a function of spike strength at locations P2 and P8. At all four locations for a spike strength of 3.5, one or more of the algorithms are able to produce an ROC curve that has an area within 2 or 3% of the maximum possible value of 1.0, which is perfect discrimination of events from background. However, events of this large magnitude are not common, and the outstanding challenge in change detection is to discriminate smaller events representative of contamination scenarios from changes of a similar magnitude caused by normal distribution system operations.

Additionally, the FAR rate for a PD of 0.95 for all ROC curves is calculated (Table 3). Across the three algorithms, the smallest FAR value corresponding to a PD of 0.95 at each location is indicated. The increment algorithm generally produces the lowest FAR values for the different locations and spike strengths, with the significant

Development of the receiver operating characteristic curve for a particular location allows a distribution system analyst to set the threshold of the detection algorithm to achieve an acceptable probability of detection.

exception of location P8 where the linear filter algorithm produces the lowest FAR values for all spike strengths. The FAR is the number of FPs divided by the number of events that actually fall below the threshold, δ_c . The number below the threshold is the sum of the FPs and the TNs (Table 1). The FAR never goes below 0.17 at location P8 even for the highest spike strengths; whereas it is down to only 1 or 2% at the highest spike strength for the three other locations.

As an example, if the FAR for a relatively high threshold, say 2.0, is 2% (0.02), then for every 1,000

TABLE 3 FAR values for a PD of 0.95 for each algorithm at each location for all 10 spike strengths

Spike Strength	Location											
	P2			P7			P8			P9		
	Parameter											
	INC	MV	LF	INC	MV	LF	INC	MV	LF	INC	MV	LF
1.0	0.83*	0.96	0.92	0.81*	0.89	0.99	0.97	0.95	0.94*	0.92*	0.97	0.96
1.1	0.51*	0.86	0.78	0.98	0.89	0.88*	0.87	0.88	0.83*	0.76*	0.96	0.87
1.2	0.37*	0.80	0.60	0.59*	0.87	0.82	0.83	0.82	0.74*	0.62*	0.93	0.86
1.3	0.33*	0.52	0.50	0.49*	0.83	0.72	0.72	0.64	0.50*	0.92	0.89	0.74*
1.4	0.26*	0.52	0.38	0.41*	0.67	0.59	0.67	0.47	0.36*	0.38*	0.85	0.68
1.5	0.23*	0.31	0.35	0.36*	0.59	0.49	0.66	0.41	0.29*	0.33*	0.85	0.62
2.0	0.12	0.09*	0.12	0.22*	0.33	0.26	0.49	0.59	0.15*	0.20*	0.52	0.41
2.5	0.08*	0.22	0.12	0.16	0.09*	0.20	0.43	0.72	0.12*	0.15	0.11*	0.27
3.0	0.08*	0.14	0.10	0.13	0.03*	0.21	0.33	0.81	0.17*	0.11	0.04*	0.15
3.5	0.07	0.02*	0.08	0.10	0.01*	0.14	0.33	0.86	0.17*	0.10	0.02*	0.15

FAR—false alarm rate, INC—time series increments, LF—linear filters, MV—multivariate distance, PD—probability of detection, ROC—receiver operating characteristic

*The minimum FAR at each location is indicated across the three algorithms.

measurement times, roughly 20 FP events will occur. For hourly measurements such as examined here, this works out to about one FP for every two days of observation. Although a 2% FAR is relatively low, depending on how these are dealt with in a practical manner by the distribution system analyst, further evaluation of these events to prove they are not TP events could still be a considerable burden. One practical option for utilities may be to decrease the sampling interval to values less than the 1-h time step used here and then build evidence of an event by combining results from consecutive time steps to identify events rather than relying on results at a single time step (Hart et al, 2007; McKenna et al, 2007). However, in the desire to identify very rare or infrequent contamination events, the FPs, not the actual detections, will always account for the vast majority of alarms in the system (Rizak & Hruday, 2006).

CONCLUSIONS

Any evaluation of the ability of a change-detection algorithm to detect anomalous events in a water distribution system must acknowledge the tradeoff between detected events and the number of FP events. Simply recording the number of correct detections is only half of the story. Here, the ROC curve is presented as an objective approach to evaluating the results of water quality change-detection algorithms on a common basis of comparison.

Three approaches to water quality change detection were devised using time series analysis tools developed in other fields. These algorithms were selected with the goals of producing accurate discrimination between normal and anomalous water quality conditions while simultaneously keeping the complexity and number of free parameters in the algorithms to a minimum. All three algorithms were tested using four water quality data sets obtained from a utility. Each data set consisted of chlorine, pH, temperature, and EC measurements made with a 1-h sampling interval. Simulated water quality events of single time step duration were added to the observed water quality data, and the strength of these events was varied across 10 levels.

Testing results showed that the three algorithms can produce ROC curve areas that are nearly 1.0 when the spike strength reaches 3.5 standard deviations. However, across all four locations and all 10 spike strengths, the multivariate distance algorithm produces the maximum area under the ROC curve in 23 of the 40 cases, with the linear filter algorithm producing all but one of the other maximum area values (Table 2). The increment algorithm only produces the maximum area in one case—for a spike strength of 1.0 when there are no events and the result is due only to chance. The performance of a single algorithm at different monitoring locations in the utility varies, and these differences appear to

be because of the nature of the variation in background water quality between locations. In some cases, two different algorithms produced areas under the ROC curves that varied by only 1 or 2%. The possible significance of these variations has not yet been examined. Future work will examine ways to determine the significance between two ROC curve areas (DeLong et al, 1988) and apply these techniques to analysis of water quality change detection.

As noted, the ROC curve area is not a unique measure of algorithm quality, and therefore the FAR at a PD of 0.95 is also evaluated as a performance measure. Results in Table 2 show that the increment algorithm produces the smallest FAR rate in 20 of the 40 cases evaluated. The MV distance algorithm and the LF algorithm produce the smallest FAR values in eight and 12 of the 40 cases, respectively. The INC algorithm is especially effective in producing the lowest FAR rates for the smaller spike strength cases.

The work presented here shows that water quality change-detection algorithms can be developed to provide an accurate indication of water quality changes from on-line water quality sensor data. However, these approaches are not perfect—there is always a nonzero number of FP detections, and these will likely be the source of most alarms when the CWS is designed to identify extremely rare true events. Additional lines of evidence beyond the change-detection algorithm must be brought to bear when separating these FP results from the TP results. These algorithms are being expanded to address the detection and characterization of water quality events that last beyond a single time step where detection of an anomalous condition for multiple time steps in a row may be used as additional evidence to better discriminate between TPs and FPs.

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